

PROBLEMS SUBMIT STATUS STANDINGS CUSTOM TEST

D. Dr. Evil Underscores

time limit per test: 1 second
memory limit per test: 256 megabytes

Today, as a friendship gift, Bakry gave Badawy n integers $a_1, a_2, \dots, a_n$ and challenged him to choose an integer X such that the value $\max_{1 \leq i \leq n} (a_i \oplus X)$ is minimum possible, where $\oplus$ denotes the bitwise XOR operation.

As always, Badawy is too lazy, so you decided to help him and find the minimum possible value of $\max_{1 \leq i \leq n} (a_i \oplus X)$.

Input

The first line contains integer n ($1 \leq n \leq 10^5$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 2^{30} - 1$).

Output

Print one integer — the minimum possible value of $\max_{1 \leq i \leq n} (a_i \oplus X)$.

Examples

input	Copy
3 1 2 3	
output	Copy
2	

input	Copy
2 1 5	
output	Copy
4	

Note

In the first sample, we can choose $X = 3$.

In the second sample, we can choose $X = 5$.

Codeforces Round 613 (Div. 2)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
282280095	Sep/21/2024 18:17	Accepted
282273149	Sep/21/2024 18:12	Wrong answer on test 1
266428908	Jun/19/2024 12:58	Wrong answer on test 3
266428625	Jun/19/2024 12:56	Wrong answer on test 1
266395102	Jun/19/2024 07:35	Wrong answer on test 1

→ Problem tags

bitmasks brute force dfs and similar divide and conquer dp greedy strings trees *1900

No tag edit access

→ Contest materials

- Announcement (en)
- Tutorial (en)